

Rakesh Kuratti

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SUMMARY

Java Backend Engineer with 5+ years of experience building and optimizing backend services using **Java, Spring Boot, REST APIs, and SQL**, with hands-on experience designing a **microservices e-commerce platform** using **Apache Kafka, Eureka, Spring Cloud Gateway, Resilience4j, JWT, PostgreSQL, and Docker**. Experienced in concurrent processing, distributed service communication, API security, event-driven architecture, fault tolerance, transaction management, and backend performance optimization.

EXPERIENCE

Infinite Computer Solutions

Dec 2023 – Sept 2025

Senior Software Engineer (Client: Nokia R&D)

Bangalore

- Reduced Optical Network Management API processing time by **40%** by parallelizing independent downstream service calls using **CompletableFuture** and **ExecutorService**, improving concurrent backend request processing
- Extended distributed backend services to support newly introduced **800 GHz Network Elements**, enhancing REST API capabilities for next-generation optical network management workflows
- Improved REST API efficiency by refactoring response DTOs to eliminate redundant data, producing leaner payloads and reducing unnecessary data transfer between distributed services
- Improved backend service maintainability by applying **object-oriented design, separation of concerns, exception handling, and reusable service components** across REST API implementations
- Improved production issue resolution by analyzing distributed service interactions, API failures, and downstream dependencies during feature development and defect fixes

Profinch Solutions Pvt. Ltd.

Dec 2022 – Oct 2023

Software Engineer (Client: UBCI Bank, Tunisia)

Bangalore

- Extended banking microservices with new loan management and financial reporting capabilities using **Java, Spring Boot, REST APIs, Hibernate, and SQL**, enabling additional business workflows
- Improved API reliability by implementing **Bean Validation**, custom business-rule validation, and centralized exception handling with **@ControllerAdvice**, providing consistent validation and error responses across REST endpoints
- Improved batch-processing throughput by parallelizing independent record-processing workloads using **Java multithreading**, reducing sequential processing overhead for large backend workloads
- Improved persistence-layer efficiency by optimizing **Hibernate/JPA database interactions, SQL queries, and entity mappings**, reducing unnecessary database processing in banking workflows
- Improved backend maintainability by applying **SOLID principles, OOP, design patterns, layered architecture, and reusable service components** while implementing new microservice functionality

TCS India Pvt. Ltd.

Nov 2020 – Dec 2022

Assistant System Engineer (Client: Mylan)

Bangalore

- Reduced unnecessary database workload by eliminating redundant database calls from backend services, improving request-processing efficiency and database utilization
- Improved large-scale record-processing efficiency by applying **Java Streams** for filtering, transformation, and aggregation operations across backend data workflows
- Delivered REST APIs end-to-end across **implementation, unit testing, API documentation, defect resolution, and deployment support**, improving reliability of production releases
- Improved backend maintainability by applying **Java OOP, exception handling, reusable service components, and separation of concerns** across application-layer implementations
- Optimized backend processing through **Java multithreading and concurrent execution** for independent workloads, improving throughput of data-processing operations

PROJECTS

E-Commerce Microservices Platform | *Java, Spring Boot, Kafka, PostgreSQL, Docker, Resilience4j* GitHub

- Designed and built a **7-service Spring Boot microservices architecture** using **Eureka service discovery, Spring Cloud Gateway, REST APIs, PostgreSQL, and Docker**, separating User, Product, Order, Payment, and Notification business domains
- Implemented **event-driven order processing** using **Apache Kafka**, publishing order events asynchronously to decouple NotificationService from the synchronous order workflow
- Prevented duplicate event processing by implementing **idempotent Kafka consumers** using event identifiers and notification-channel checks, ensuring repeated message delivery does not trigger duplicate notifications
- Improved distributed-system resilience by configuring **Resilience4j circuit breakers, failure-rate thresholds, timeouts, and fallback handling** for inter-service REST communication, preventing cascading downstream failures
- Secured microservices using **Spring Security, JWT authentication, BCrypt password hashing, stateless sessions, and role-based access control**, independently validating authenticated requests across service boundaries
- Implemented transactional order and inventory workflows using **Spring Data JPA, Hibernate, PostgreSQL, @Transactional, validation, and domain-specific exception handling**, maintaining consistency across core e-commerce operations

EDUCATION

KLE Technological University
Bachelor of Engineering in Computer Science

Aug 2017 – May 2020
Hubli, Karnataka

SKILLS

Languages : Java, SQL

Frameworks : Spring Boot, Spring Security, Spring Data JPA, Hibernate, Spring Cloud Gateway, Eureka

Distributed Systems : Microservices, Service Discovery, API Gateway, Inter-Service Communication, Fault Tolerance, Circuit Breaker, Event-Driven Architecture

Messaging : Apache Kafka, Kafka Producers, Kafka Consumers, Idempotent Consumers, Asynchronous Processing

Security : JWT, Spring Security, Role-Based Access Control, BCrypt, Stateless Authentication

Backend : REST APIs, DTOs, Bean Validation, Exception Handling, @ControllerAdvice

Resilience : Resilience4j, Circuit Breaker, Timeout, Fallback, Failure Rate Threshold

Persistence : PostgreSQL, MySQL, Spring Data JPA, Hibernate, JDBC, SQL

Concurrency : Multithreading, ExecutorService, CompletableFuture, Java Streams, Concurrent Processing

Architecture : Microservices, Database-per-Service, Synchronous REST, Asynchronous Messaging, Layered Architecture

Tools : Git, Docker, Maven, Jira, Confluence, Swagger/OpenAPI, Spring Boot Actuator

Concepts : OOP, Data Structures and Algorithms, Design Patterns, SOLID Principles, System Design, Transaction Management